An AI-assisted investigation of a Spin(10) × global $U(1)_{PQ}$ gauge/scalar candidate

Construction, bounded results, and failed tests

Solo research discussion note, prepared with AI assistance. Version 4.1.

Abstract

This note specifies a nonsupersymmetric Spin(10) gauge/scalar construction with global PQ selection rules. Earlier scalar points include a failed Higgs quartic tuning and a positive-quartic point subsequently rejected by a tree perturbativity screen. A different exploratory point has positive heavy masses, a positive relaxed Higgs quartic, global quartic positivity and a complete scalar-contact bound below 1/2. Its local-inclusive one-loop singlet pole shift is large; newer subtraction-independent momentum combinations give distinct, smaller diagnostics without proving quantum control. Formal one-loop UV/normalization results and four scoped two-loop primitive amplitude families have now been calculated. The complete finite gauge threshold, unification verdict and physical viability remain unresolved. All results retain their point, reference, loop-order and domain restrictions. Publication makes no new scientific acceptance decision.

1. Assumptions and fields

Work in four spacetime dimensions with gauge group Spin(10) and global PQ selection rules. The simply connected group permits the fermionic spinor 16; local tensor calculations use the $\mathfrak{so}(10)$ action. A possible quotient by common gauge/PQ centers has not been fixed, although the ordinary-field center identification has now been calculated (Section 5). No supersymmetry, explicit soft PQ violation, or additional scalar representation is assumed.

Field	Representation/type	q_{PQ}	Real scalar coordinates
Φ_{ij}	real symmetric traceless 54	0	54
Σ_{ijklm}	complex self-dual 126	+2	252
ϕ_i	complex 10	−2	20
S	complex singlet	−4	2
ψ_p, $p = 1, 2, 3$	left-handed 16_F	+1	—

Indices run from 0 to 9, with Euclidean vector metric δ_{ij} and $\epsilon_{0\cdots9} = +1$. Fix $*\Sigma = +i\Sigma$; its conjugate is the $\overline{126}$. The scalar kinetic convention is

$$\mathcal{L}_{\rm kin} = \frac{1}{2}(D_\mu\Phi_{ij})^*(D^\mu\Phi_{ij}) + \frac{1}{2\,5!}(D_\mu\Sigma_{ijklm})^*(D^\mu\Sigma_{ijklm}) + (D_\mu\phi_i)^*(D^\mu\phi_i) + (\partial_\mu S)^*(\partial^\mu S).$$

The self-dual factor avoids double counting complementary five-index sectors. The physical construction uses the usual gauge-curvature kinetic term and $\mathcal{L} = \mathcal{L}_{\rm gauge} + \mathcal{L}_{\rm kin} + \mathcal{L}_Y - V$. The allowed symbolic Yukawa structures are $\psi\psi\phi$ and $\psi\psi\Sigma^*$, with complex symmetric family matrices and isometrically normalized intertwiners. Their numerical matrices and right-handed-neutrino spectrum have not been selected.

The field-content motivation follows Babu and Khan [1]; PQ/axion context is discussed in [2,3]. The potential below is a separately specified source-derived variant. Published scalar benchmarks, fermion fits, and unification conclusions are not imported as results for it. No novelty claim is made.

2. Complete frozen scalar definition

Write $V = V_2 + V_3 + V_4$. Repeated tensor indices are summed over all ten values. Define

$$N = \frac{1}{5!}\Sigma_{abcde}\Sigma^*_{abcde}, \quad u = \phi_i^*\phi_i, \quad t = \phi_i\phi_i, \quad p_n = \mathrm{Tr}(\Phi^n),\ n = 2,3,4,$$

$$K_{ij} = \frac{1}{4!}\Sigma_{iabcd}\Sigma^*_{jabcd}, \qquad T_{ij} = \frac{1}{4!}\Sigma_{iabcd}\Sigma_{jabcd},$$

$$L_{\Phi\Sigma} = \frac{1}{2\,3!}\Phi_{ij}\Phi_{kl}\Sigma_{ikabc}\Sigma^*_{jlabc}, \qquad E = \frac{1}{2!\,2!\,3!}\Sigma_{ijlmn}\Sigma_{ijpqr}\Sigma^*_{lmpqr}\phi_n.$$

T is symmetric traceless. For increasing tuples I, J of length $5-k$, let $P_k[I,J] = \sum_{K\in\binom{\{0,\ldots,9\}}{k}} \Sigma_{KI}\Sigma_{KJ}$ and $Q_k = \sum_{I,J}|P_k[I,J]|^2$, for $k = 0,1,2$. Antisymmetric components with repeated indices vanish. If A, B are disjoint, $\varepsilon(A,B)$ is the parity of their concatenation relative to increasing order; otherwise it is zero. The fourth real five-form quartic is

$$X_{131} = \sum_{I,J\in\binom{\{0,\ldots,9\}}{4}} \sum_{\substack{A\subset I,\ |A|=3\\ B\subset J,\ |B|=1}} \varepsilon(A, I\setminus A)\varepsilon(B, J\setminus B)\varepsilon(A,B)\varepsilon(I\setminus A, J\setminus B)$$
$$\times P_1[I,J]\,\overline{P_1[\mathrm{sort}(A\cup B), \mathrm{sort}((I\setminus A)\cup(J\setminus B))]}.$$

The potential in these normalizations is

$$\begin{aligned}
V_2 &= m_\Phi^2 p_2 + m_\Sigma^2 N + m_\phi^2 u + m_S^2|S|^2,\\
V_3 &= \mu_\Phi p_3 + \mu_{\Phi\phi}\Phi_{ij}\phi_i^*\phi_j + [z_6 t S^* + \text{h.c.}],\\
V_4 &= \lambda_{\Phi1}p_2^2 + \lambda_{\Phi2}p_4 + \lambda_{\Phi\Sigma1}p_2 N + \lambda_{\Phi\Sigma2}L_{\Phi\Sigma}\\
&\quad + \lambda_{\Phi\phi1}p_2 u + \lambda_{\Phi\phi2}\phi_i^*\Phi_{ij}\Phi_{jk}\phi_k + \lambda_{\Phi S}p_2|S|^2\\
&\quad + \lambda_{\Sigma1}Q_0 + \lambda_{\Sigma2}Q_1 + \lambda_{\Sigma3}Q_2 + \lambda_{\Sigma4}X_{131}\\
&\quad + \lambda_{\Sigma\phi1}N u + \lambda_{\Sigma\phi2}\phi_i^*K_{ij}\phi_j + \lambda_{\phi1}u^2 + \lambda_{\phi2}|t|^2\\
&\quad + \lambda_{\Sigma S}N|S|^2 + \lambda_{\phi S}u|S|^2 + \lambda_S|S|^4\\
&\quad + [z_4\Phi_{ij}T_{ij}S + z_K\Phi_{ij}\phi_i\phi_j S^* + z_\eta E + z_D T_{ij}\phi_i\phi_j + \text{h.c.}].
\end{aligned}$$

All m^2, μ, λ coefficients are real; $z_6, z_4, z_K, z_\eta, z_D$ are complex. The masses have dimension two, the cubics dimension one, and the quartics dimension zero. Hermitian conjugation includes the coefficient. The shorthand $\mu_{\Phi\phi}, \lambda_{\phi1,2}, \lambda_{\phi S}, z_\eta$ corresponds to repository names `muPhiPhi`, `lambdaPhiVector1,2`, `lambdaVectorS`, and `zEta`.

There are 29 Hermitian coefficient groups: four quadratic, three cubic, and 22 quartic. Their 24 real and five complex coefficients give 34 real directions before rephasings. Independent D_5 character multiplicities give 34 complex singlet slots before Hermitian pairing; explicit contractions attain that upper bound. The generic scalar phase quotient leaves three relative coupling phases. This basis result fixes the action, not a viable vacuum.

3. Scalar results: a failed tuning and a later positive point

The candidate high-scale background is

$$\langle\Phi\rangle = \frac{\omega}{\sqrt{60}}\,\mathrm{diag}(-2\mathbf{1}_6, 3\mathbf{1}_4), \quad \langle\Sigma\rangle = \frac{\sigma}{4\sqrt{2}}\bigwedge_{k=0}^{4}(e_{2k+1}+ie_{2k}), \quad \langle S\rangle = \frac{v}{\sqrt{2}}, \quad \langle\phi\rangle = 0.$$

The first direction has the intended Pati–Salam/discrete-parity stabilizer; the combined gauge orbit has 33 broken and 12 unbroken directions, consistent with the SM Lie algebra. Global topology and the PQ quotient are separate questions. Tadpoles and Hessians are derived from the same frozen potential.

At the recorded Stage 1 point, $\omega = \sqrt{60}$ in calculation units and $\sigma/\omega = v/\omega = 0.1$. The exact-decimal free coefficients and tadpole-eliminated masses are specified in `STAGE1_POINT.json` and its replay program. All 35 SM-irrep multiplicity blocks were evaluated. The real Hessian has rank 294 and nullity 34: 33 gauge directions plus one PQ direction. The remaining eigenvalues are positive in the canonical kinetic metric; the smallest recorded physical mass squared is approximately 0.08656. This is local positivity modulo symmetry, not global boundedness or a phenomenologically fitted scale. This point has no light Higgs doublet.

The distinct Stage 2 point changes $\mathrm{Re}\,z_6, \mathrm{Re}\,z_K, \mathrm{Re}\,z_\eta$ to 0.3 in the stored dimensionless convention and tunes $m_\phi^2/\omega^2 \simeq -0.1337864654$ to the exact algebraic root recorded in `STAGE2_POINT.json`. It has rank 290 and nullity 38, including four real coordinates of one massless complex SM Higgs doublet. All physical heavy quadratic modes remain positive. This statement alone is insufficient for local stability along the tuned flat direction.

Decisive additional test

For a canonically normalized neutral field h, $H^\dagger H = h^2/2$, expand around the tuned point with physical heavy coordinates X_A:

$$V = V_0 + a_{\mathrm{direct}}h^4 + h^2 j_A X_A + \tfrac{1}{2}X_A\mathcal{H}_{AB}X_B + \cdots .$$

Minimization on the positive heavy subspace gives $X = -\mathcal{H}^{+}jh^2 + \cdots$, and hence

$$\lambda_{\mathrm{eff}} = 4\left(a_{\mathrm{direct}} - \tfrac{1}{2}j^T\mathcal{H}^{+}j\right), \qquad V_{\mathrm{eff}} = \lambda_{\mathrm{eff}}(H^\dagger H)^2 + \cdots .$$

The recorded parent-action evaluation gives

Quantity in $(H^\dagger H)^2$ normalization	Value
Fixed-heavy direct quartic	+0.19910
Heavy relaxation subtraction	56.68893
Relaxed effective quartic	**−56.48984**

The negative sign makes the Stage 2 point a failed local scalar minimum at tree level. It does not invalidate the separate Stage 1 result or prove that every parameter point fails. A direct parent-potential expansion along the heavy-field valley reproduces the quartic; heavy-coordinate shear and complex Higgs-phase controls preserve it. These extraction routes share the frozen tensor implementation. No global-minimum or loop-corrected stabilization claim is made.

A distinct positive-quartic point

The later recorded point in `canonical_so10_positive_higgs/POINT.json` retains the Stage 1 VEVs but uses $z_6/\omega = z_K = 0.03$, $z_\eta = 0.3$, $\lambda_{\Phi\phi 1} = -0.002$ and an exact algebraic tuning $m_\phi^2/\omega^2 \simeq 0.007302047861$. Its real Hessian has rank 290 and nullity 38, with positive heavy modes. The direct quartic is 0.19845235; heavy relaxation subtracts 0.09359411, leaving

$$\boxed{\lambda_{\mathrm{eff}} = +0.10485824.}$$

The point's SHA-256 is `476645438ea8a7ada42fea90e80174b772d45ce3e729d013d134e9848950c816`. Direct-parent and valley-polynomial routes agree with shared tensor inputs. This remains a local tree-level result. Subsequent calculation at this SAME point gives a physical coherent scalar channel with $|a_0| = 0.76809032 > 1/2$ in the fixed-input high-energy tree limit [7]. Its vector-exchange currents vanish in that channel. The point therefore fails this perturbativity screen; positive local masses and quartic did not establish a controlled benchmark. This failure does not exclude the construction. The new exploratory point in Section 6 has different inputs and must be assessed separately.

4. Gauge matching: a calculated mismatch, an incomplete verdict

The recorded direct one-loop calculation matches the full parent to the SM in a common $\overline{\mathrm{MS}}$ background-field Feynman-gauge convention, with GUT-normalized g_1. It includes 290 heavy real scalar directions and 33 massive vector/Goldstone/ghost sets from the later positive-quartic point, not the earlier failed tuning. The threshold ledger binds that point's hash. This point subsequently fails the scalar perturbativity screen. These residuals have not been recomputed for the new exploratory point in Section 6.

Define $\alpha_i = g_i^2/(4\pi)$ and one-loop threshold functions λ_i by

$$\alpha_i^{-1}(\mu) = \alpha_{10}^{-1}(\mu) - \frac{\lambda_i(\mu)}{12\pi}.$$

The complete state accounting gives

$$b_{\mathrm{SM}} = \left(\frac{41}{10}, -\frac{19}{6}, -7\right), \quad b_S = \left(\frac{377}{30}, \frac{77}{6}, \frac{79}{6}\right), \quad T_V = (8, 6, 5),$$
$$b_{\mathrm{SM},i} + b_{S,i} - \frac{7}{2}T_{V,i} = -\frac{34}{3} = b_{10}, \qquad \frac{d\lambda_i}{d\ln\mu} = 6(b_{\mathrm{SM},i} - b_{10}).$$

These are exact beta-jump and one-loop matching-scale checks. At $\mu = \omega$, the frozen inverse-coupling inputs are $(59.0100, 29.5800, 8.47458)$ at M_Z. The one-loop equations are

$$\alpha_i^{-1}(M_Z) = \frac{4\pi}{g_{10}^2} - \frac{\lambda_i^S + \lambda_i^V(g_{10})}{12\pi} + \frac{b_i^{\mathrm{SM}}}{2\pi}\ln\frac{\omega}{M_Z}.$$

They have no common perturbative solution for the frozen inputs. The recorded least-squares residuals, in inverse-coupling units and channel order $(1, 2, 3)$, are:

Completed approximation	r_1	r_2	r_3	$\max\lvert r_i\rvert$
One-loop running and thresholds	0.36615	−1.05980	0.69389	1.05980
Two-loop SM running, one-loop thresholds	0.26091	−0.74548	0.48479	0.74548

The latter uses the SM running framework [5] and frozen reference boundaries from [6], including top, bottom, and tau insertions with their one-loop evolution. A separate inverse-coupling evolution with a different adaptive integrator reproduces the residual to better than 2×10^{-8} under the frozen equations. Matching-scale variation $\mu/\omega \in [0.5, 2]$ leaves maximum residuals from 0.74172 to 0.74931.

What the missing calculation must supply

The complete finite two-loop heavy boundary is still unavailable. Beyond the earlier RG logarithmic template, explicit model-specific primitive contributions have now been calculated in the same older point's fixed reference (Section 7). They have not been combined into a full renormalized threshold or a new unification refit. The published general matching framework [4] provides methodology and a restricted comparator, not the finite answer for these mixed, nondegenerate fields and multiple VEVs. The complete Yukawa-dependent threshold also requires numerical family matrices and right-handed-neutrino masses.

Small scale variation is not a rigorous bound on unknown finite constants. Setting them to zero or freely tuning them would not compute a prediction. Consequently the formal gauge verdict remains unresolved: neither unification success nor exclusion has been established. The diagnostic fitted $g_{10} \simeq 0.5231$ and $\omega \simeq 1.64 \times 10^{15}\,\mathrm{GeV}$ are not physical benchmark parameters. No replacement spectrum or flavor fit is introduced in this note.

5. Symmetry and quantum evidence: expanded, still scoped

The separately named covariant gauge/scalar candidate satisfies the recorded classical local identities through the 328-real scalar representation and all 45 generators. It is an inactive qualification, not replacement of a promoted component action. Earlier literal-action failures remain preserved; one has

$$\delta\mathcal{L}_{\text{old}} = \frac{ig_{10}}{\sqrt{2}}(\partial_\mu C^4)(Y\partial^\mu X - X\partial^\mu Y) \neq 0.$$

The later candidate repairs that kinetic structure by derivation. A failed encoding does not refute Spin(10). Likewise the preserved auxiliary-cubic normalization counterexample, in common $\hbar g_{10}\xi^2/(32\pi^2\epsilon)$ units, remains

$$G_{\text{actual}} = \frac{48\sqrt{10}}{25}, \quad G_{\text{candidate}} = \frac{8\sqrt{10}}{5}, \quad G_{\text{actual}} - G_{\text{candidate}} = \frac{8\sqrt{10}}{25}.$$

Formal one-loop progress

The earlier scalar-pole closure has rank $4 + 4 + 26 = 34$ within its tested operator space. Subsequent work goes further: the complete formal local one-loop logarithmic residue in the explicitly ordered mixed-reference/heat family obeys $Q\Gamma_{1,\log} = 0$ as an integrated local functional. Its proof retains the source Euler terms, reference subtraction and boundary currents. Cyclicity is used for the integrated homogeneous UV residue, not for an arbitrary finite heat-weighted trace. This is not an identity for the complete finite physical effective action.

The actual parity-even bosonic counterterm has also been reconstructed within 34 real scalar-potential directions, four scalar kinetic metrics and one Yang–Mills metric, plus separate background/vacuum normalizations. These are operator-coordinate counts, not changes to the formal 34/39 authority ledger. Correlated first-order exact-term normalization and physical parameter closure are qualified on the stated coefficient domain. A stricter higher-order extension cannot be imposed retroactively as a missing one-loop coefficient test; the broader original quantum acceptance obligations still remain.

Constructed graded causal products now support a local anomalous quantum action principle at finite order on a compactly switched, off-shell domain [11,12].

The required heat-to-causal finite conversion, simultaneous Slavnov/background normalization and an explicit relative reference member have been retained. This improves on the earlier merely conditional locality discussion. It neither removes the insertion of a classical switching breaking nor proves a global interacting vacuum, physical infrared limit or unique historical prescription. An actual source-quadratic counterexample shows that agreement of relative field products alone cannot determine the full retained-source functional. Full E5/A5 and physical quantum equivalence remain unqualified; the formal state is still version 113, at 34/39.

Gauge anomaly and PQ structure

For the three left-handed 16_F multiplets, the local four-dimensional perturbative gauge-anomaly tensor and linear gauge traces vanish. A weight polynomial route with conjugacy/polarization and a Clifford-matrix route evaluate the established representation criterion [9], including all 16,215 symmetric cubic slots. This does not construct a regulated matter determinant or settle global anomalies, bundles or the full quantum measure.

For the declared PQ charges the convention $N = 2\sum_f q_f T_c(f)$ and $E = 2\sum_f q_f Q_f^2$ gives

$$N = 12, \qquad E = 32, \qquad E/N = 8/3.$$

The common Spin(10) center identifies an ordinary-field PQ quarter-turn; nonzero gauge-singlet S fixes the vacuum-orbit period to $\alpha = \pi/2$. The conventional QCD potential would consequently give $N_{\text{DW}} = 3$, *conditional* on realistic nonzero quark masses/QCD susceptibility, this ordinary orbit and no additional identification or explicit PQ breaking [3]. The electroweak-unbroken reference does not supply that QCD vacuum. This is neither a physical axion mass/abundance prediction nor a full global completion.

6. A new scalar witness and its quantum-control warning

The different rational fixture in `canonical_so10_coherent_joint_witness/POINT.json` uses the same field content, action and VEV shape. Its frozen reduced inputs are $p_1 = 1/100$, $p_2 = 1/50$, $x = 3/25$, $s = 1/1000$, $z = 1/500$, $D = 1/50$, $t = 23/1250$, $\Delta = 0$, $\eta = 1/10000$; the dependent coefficients and larger simple weak root are fixed by the recorded action maps. It is an exploratory fixture, not an adopted physical benchmark. Do not combine its results with the old point's spectrum, thresholds or loops.

Exact specialization of the inherited 12 spectral types (35 blocks, 328 real fields) gives 290 positive directions and 38 symmetry/tuning zeros. The actual weak kernel and all positive neutral relaxation modes give $\lambda_{\rm eff} \simeq 0.1074374161 > 0$ at tree level. This uses a new point-specific light direction, not a transported earlier quartic.

Complete classical coverage at this point

The complete symmetric scalar pair space has 53,956 states. In the contact convention of [7], the recorded structural Gram and block-norm bounds give

$$\|a_0^{\rm contact}\| \le \frac{35439836953}{240000000000\pi} \simeq 0.47003544 < \frac{1}{2}.$$

This is a conservative upper bound, not a largest eigenvalue or complete gauged scattering amplitude. Massless-gauge exchange requires its own infrared-safe treatment; no gauge coupling is silently set to zero.

With $X = \operatorname{Tr}\Phi^2$, $n = N_\Sigma$, $u = \phi^\dagger\phi$, $v = |S|^2$, global invariant inequalities and explicit completion of the complex mixed terms give a rational positive-definite four-norm lower matrix. Its exact minors and a separate rational solver yield, for all 328 real fields,

$$V_4 \ge c(X + n + 2u + 2v)^2, \qquad c = \frac{82440401071117326624733394 9}{17205798939483426705901101207 50} \simeq 4.79143115 \times 10^{-4} > 0.$$

This is global quartic positivity, rather than sampled positive rays. It ensures growth at large fields for this potential. It does not prove that the chosen stationary vacuum is deepest or has a sufficient tunneling lifetime. The old failed points and closed parameter families remain failed in their qualified scopes; these certificates do not transfer to other inputs.

A complete selected one-loop pole coefficient is large

At the same fixture, $Q^2 = \omega^2 = 60$ and the declared $\overline{\rm MS}$ stationary-branch expansion, the isolated singlet has tree squared mass $6/5$. Scalar seagulls and momentum-dependent bubbles, including the physical PQ/Higgs zero modes, give a one-particle-irreducible contribution $-1.36799138 - 0.23557167i$. The required vacuum-source insertion adds $+3.03144395$, so [8]

$$\delta m^2_{S,\rm pole} = 1.66345257 - 0.23557167i, \qquad \frac{|\delta m^2_{S,\rm pole}|}{m^2_{S,\rm tree}} \simeq 1.40004.$$

Field-content and projector checks justify this particular one-loop component. Direct singlet gauge/Yukawa vertices vanish here. The tree singlet plane is isolated and the relevant other tadpole multipliers vanish. This does not make every quantum-vacuum component complete. An exact local RG/log check retains generated Φ–S and Σ–S portals [10] and cancels the scale derivative, but does not reduce the large correction. A small first-order correction is unsupported in this expansion. No reliable physical pole mass/width, resummed solution or universal scheme failure is inferred. Classical feasibility and controlled quantum viability remain distinct.

7. New physical diagnostics and finite-threshold progress

The same singlet has distinct momentum diagnostics

At the exploratory point of Section 6, let $a = m^2_{S,\mathrm{tree}} = 6/5$ and $\Pi(s)$ be its complete retained one-loop self-energy. New combinations give

$$\begin{aligned} E &= \Pi(-a) - 2\Pi(-2a) + \Pi(-3a), & E/a &\simeq -0.01940549, \\ T &= \Pi(a) - 3\Pi(-a) + 2\Pi(-2a), & T/a &\simeq 0.05035452 - 0.19630972i, \\ & & |T|/a &\simeq 0.20266496. \end{aligned}$$

Both cancel every local affine function $c_0 + c_1 s$, including mass and wavefunction subtractions, seagulls and the vacuum-source constant at this order. Their magnitudes are about 1.94% and 20.27% of the tree squared mass. They do not replace or invalidate the earlier 140% local-inclusive pole shift; the absorptive cut remains. Neither a selected small combination nor the large pole shift alone supplies a higher-order error bound.

The absence of a direct S Yukawa vertex at one loop does not close the two-loop singlet problem. An actual scalar/fermion topology has a nonzero unweighted carrier $1.1880059743\,\mathrm{Tr}(Y_{10}^\dagger Y_{10})$ in the recorded units. This is graph support, not an integrated correction. Unfrozen Yukawa matrices and masses remain inputs. Uniform scalar-potential weakening preserves the proven classical properties, but worsens the tested fixed-gauge bosonic radial response; it is not a demonstrated rescue.

Four of nine primitive amplitude families now have scoped results

The separate C1 background-field calculation uses the *older* +0.10485824 point, hash `47664543...50c816`, in its declared $\overline{\mathrm{MS}}$ background-field Feynman-gauge reference, $\xi = 1$. It is not the 643-field formal nonlinear-source construction and does not transfer the Section 6 point's spectrum or loop numbers. The old point's perturbativity failure is retained.

Primitive	Completed scope and boundary
VVVV	Actual quartic-vector bare F^2 coefficient and complete regulated infrared/EFT subtraction; UV subdivergences and poles remain.
$VVSS$	Mixed quartic-vector/scalar F^2 coefficient with graph-local UV subtraction and actual light matching.
VSS	All five vector mass sectors and all 328 scalar directions, including mixing and soft modes, assembled at one common scale; overall UV poles remain.
ghost–S	Ordered ghost charges, full scalar spectral measure and finite primitive with proper forests and one-soft matching; no fitted counterterm.

Here V is a quantum vector and S in the graph labels denotes a scalar line of any allowed species, not specifically the singlet field. The count is an inventory, not a completion percentage or a count of fully renormalized physical thresholds. Complete $SSSS$, VVV, VVS, SSS and ghost–V families remain, alongside the parent-minus-family counterterm remainder, overall two-loop background counterterm, finite field/parameter/reference/tadpole/width insertions, tree/EFT matching and the nonlinear conversion to $C_{1,\mathrm{GS}}$. Missing terms are not set to zero. Diagnostic $g_{10}^2 = 1/2, 1$ are not adopted physical inputs. No finite C1 value or updated unification fit is claimed.

8. Evidence and reproduction map

The frozen source checkpoint is

`7562f42c14ef1ab5a946fbfc4534509f3378c76b`.

Source `project_state.json` remains version 113, unchanged by publication. The original public Versions 2 and 3 and all v1/v2 source archives/manifests are preserved; the v3 source update overlays them. The reproduction guide specifies exact assembly and separate historical-policy and external-control namespaces. Original Git history is absent; custody exit 2/seven unresolved references is not PASS.

The following folders are under `calculations/`, except the last row. Read the frozen contract, actual completed receipt, inputs, derivation and successful/failed captures together. Earlier reports retain their chronological limits; later certificates do not rewrite them.

Scope	Folder locator
Action / old points	`canonical_so10_scalar_reconstruction/` `canonical_so10_scalar_benchmark/`
Failed / positive quartics	`canonical_so10_light_higgs_quartic/` `canonical_so10_positive_higgs/`
Old contact failure	`canonical_so10_radial_input_control/`

New fixture / bounds	`canonical_so10_coherent_joint_witness/`
	`canonical_so10_complete_contact_bound/`
	`canonical_so10_finite_full_bfb/`

Singlet pole / RG	`canonical_so10_singlet_pole_control/` `canonical_so10_singlet_pole_rg_check/`

Shape / missing input	`canonical_so10_singlet_momentum_shape/`
	`canonical_so10_singlet_two_loop_input_closure/`
	`canonical_so10_scalar_gauge_homothety/`

Historical gauge fit	`canonical_so10_direct_gauge_next_order/`

Formal UV / closure	`canonical_so10_bv_uv_residue_bridge/`
	`canonical_so10_physical_counterterm_span/`
	`canonical_so10_one_loop_acceptance_reconciliation/`

Local products / QAP	`canonical_so10_graded_causal_qap/`
	`canonical_so10_background_reference_normalization/`
	`canonical_so10_local_cutoff_comparison/`

Primitive amplitudes	`canonical_so10_vector_quartic_ir_matching/`
	`canonical_so10_vector_scalar_quartic/`
	`canonical_so10_scalar_current_vss_assembly/`
	`canonical_so10_ghost_scalar_sunset/`

Gauge / PQ structure	`canonical_so10_local_gauge_anomaly/` `canonical_so10_pq_anomaly_screen/`
Portable complete pilot	`tools/portable_dh_v1/README.md`

What readers can rerun

The portable pilot needs Python 3.11+ and the standard library. Its slot and left-derivative methods compute every coefficient of a 99,256-position classical Hessian identity before opening the reference table. It inherits the interpretation of its frozen action projections; it does not reproduce the full quantum model or upstream physical proofs. Both documented methods are run for this release in an isolated copy.

The package verifier authenticates every released file and archive member; the publication receipt separately lists receipt/input checks, actual replays and omissions. Expensive original Hessian, pole and threshold computations are not repeated for packaging. Some historical harnesses retain absolute paths and read guards; explicit relocation is needed, without editing frozen evidence or bypassing integrity checks.

9. Remaining obligations and AI assistance

No physically viable GUT or experimental agreement is established. The new point has a full classical quartic bound, but the preferred vacuum, tunneling, loop control, hierarchy/naturalness and complete gauged scattering remain open. Full quantum restoration/measure/contact/IR obligations, the complete finite heavy threshold, realistic matter/neutrino inputs, complete axion cosmology and proton-decay phenomenology also remain open. The old gauge residuals are diagnostics at their own frozen inputs and do not predict the new point.

Useful discussion questions are how to define a controlled physical-input expansion retaining the computed singlet cut, and how to finish the remaining primitive/counterterm/EFT layers without double counting. Local formal one-loop results and partial two-loop amplitudes should be assessed in their own domains. Missing terms are not assumed to rescue any failed result.

LLMs assisted derivations, code, verification and writing. Scientific direction and acceptance belong to the human author. The records demonstrate bounded symbolic/numerical calculation, structural reductions and cross-checking; they also preserve failed interpolation, assertion, normalization and verification attempts. Corrected agreement cannot validate a shared physical assumption. Shared action/representation inputs, inherited proof caches, some tensor/arithmetic code, Python/SymPy and the same analyst limit independence. Selected work uses NumPy, SciPy, mpmath, CVXPY/Clarabel and Julia. Multiple AI routes are not independent scientists. No external expert verification, whole-model proof-assistant certification or general reliability claim is made.

References

[1] K. S. Babu and S. Khan, *A Minimal Non-Supersymmetric SO(10) Model: Gauge Coupling Unification, Proton Decay and Fermion masses*, Phys. Rev. D **92**, 075018 (2015). arXiv:1507.06712.

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